AUTONOMOUS ERROR DETECTION AND SELF-HEALING
MULTI-AGENT SYSTEMS

TT3 • Module: Agent Development with AutoGen and CrewAI

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1. CONCEPT

Autonomous error detection means the system identifies abnormal, failed, inconsistent, or invalid agent behaviour without manually inspecting every step.

Self-healing means the system automatically attempts recovery—such as retrying, correcting inputs, changing tools, replanning, or restoring state—and then verifies the recovered result.

Core loop: Detect → Diagnose → Recover → Verify → Continue.

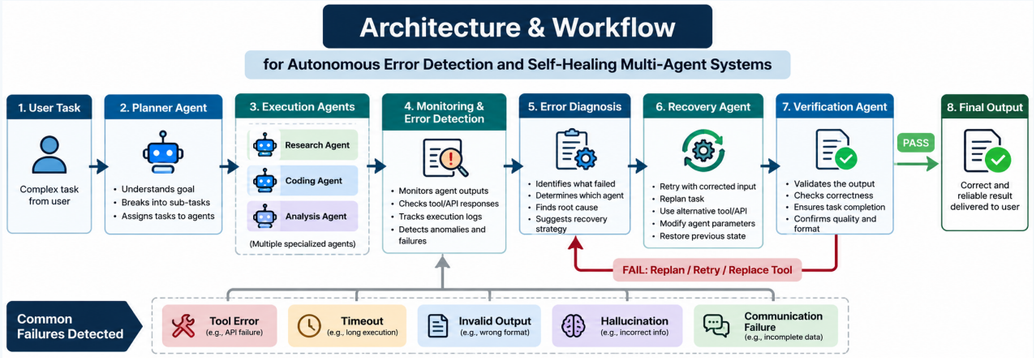
2. COMMON FAILURES

- **Tool/API failure:** service unavailable or malformed request
- **Timeout:** execution exceeds allowed time
- **Invalid output:** wrong format or missing fields
- **Hallucination:** plausible but incorrect information
- **Communication failure:** incomplete or contradictory agent messages
- **Planning failure:** incorrect task sequence or dependency

3. BENEFITS

Reliability: fewer workflow failures
Resilience: recovery from transient faults
Less manual intervention: routine failures can be handled automatically
Better task completion: validation prevents silent failures
Scalability: monitoring and recovery can be applied across many agents.

7. ARCHITECTURE & SELF-HEALING WORKFLOW



4. CHALLENGES

1. **Error propagation:** one bad result can affect downstream agents.
2. **Error localization:** finding the original faulty step is difficult.
3. **False recovery:** a retry may still produce an incorrect answer.
4. **Retry loops:** repeated recovery increases cost and latency.
5. **Safety:** autonomous changes need limits and approval for critical actions.

5. POSSIBLE SOLUTIONS

- **Output validation** between critical stages
- **Failure classification** using logs, tool responses, and consistency checks
- **Bounded retries** with timeouts and termination conditions
- **Adaptive replanning** when the original plan fails
- **Fallback tools/agents** for unavailable services
- **Human approval** for high-impact actions

6. REAL-WORLD APPLICATION

Self-healing AI coding team

Planner Agent → Coding Agent → Testing Agent → Error Detector
If tests fail, the Recovery Agent diagnoses the failure, asks the Coding Agent to revise the implementation, and the Testing Agent runs again.

Other applications: automated data pipelines, customer support, cybersecurity monitoring, financial workflows, and ML pipeline generation.

Framework connection: AutoGen GraphFlow supports structured, conditional and looping agent workflows.
CrewAI Crews support role-based collaboration; CrewAI Flows add structured, event-driven control and state.

10. OBJECTIVES

- Identify common errors in collaborative multi-agent workflows.
- Design a workflow for automatic error detection and diagnosis.
- Apply recovery strategies such as retry, replanning and tool replacement.
- Verify recovered outputs before continuing execution.
- Improve reliability, resilience and reduce unnecessary human intervention.

11. AUTOGEN & CREWAI CONNECTION

AutoGen: structured multi-agent workflows can support sequential, parallel, conditional and looping execution with termination controls.
CrewAI: role-based agents collaborate through Crews, while Flows provide structured execution, state and event-driven control.
Both can be combined with monitoring, validation and recovery logic to build more reliable workflows.

12. EVALUATION PARAMETERS

- **Error Detection Accuracy**
  How accurately failures are identified.
- **Error Detection Rate**
  Percentage of failures successfully detected.
- **Recovery Success Rate**
  Percentage of detected errors successfully recovered.
- **Task Success Rate**
  Percentage of tasks completed correctly after recovery.
- **Recovery Latency**
  Time required to detect and recover from an error.

KEY TAKEAWAY: Detect → Diagnose → Recover → Verify → Continue